THE COMPLETION OF A DIRECT SUM OF COPIES OF A NOETHERIAN RING IS FLAT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The completion map is flat, flatness is stable under composition, and completing a complete ring does not change it. Algebra, Lemma \ref{algebra-lemma-completion-flat}, Algebra, Lemma \ref{algebra-lemma-composition-flat}, and Algebra, Lemma \ref{algebra-lemma-completion-complete}, cited at proof lines 6420-6431. Justify replacing the ring by its completion and rewriting the completed direct sum.
- The comparison map from the completed direct sum into the product is universally injective. Lemma \ref{lemma-ui-completion-direct-sum-into-product}, cited at proof lines 6432-6434. Reduces flatness of the completed direct sum to flatness of the product.
- A universally injective map out of a module detects flatness. Algebra, Lemma \ref{algebra-lemma-ui-flat-domain}, cited at proof lines 6435-6436. Carries flatness of the product back to the completed direct sum.
- Products of copies of a coherent ring are flat, and Noetherian rings are coherent. Algebra, Proposition \ref{algebra-proposition-characterize-coherent} and Algebra, Lemma \ref{algebra-lemma-Noetherian-coherent}, cited at proof lines 6437-6440. Give flatness of the product of copies of the ring.

Source reference. Original lemma 06LE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8463–8463. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00MB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 817–817. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 638–638. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 031C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3979–3979. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05BC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6880–6880. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 058P: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6787–6787. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05CZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6937–6937. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05CY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6936–6936. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $I \subset R$ be an ideal. Let A be a set. Assume R is Noetherian. The completion $(\bigoplus_{\alpha\in A} R)^\wedge$ is a flat R-module.*

Proof. Denote $R^\wedge$ the completion of R with respect to I. As $R \to R^\wedge$ is flat by Algebra, Lemma 00MB it suffices to prove that $(\bigoplus_{\alpha\in A} R)^\wedge$ is a flat $R^\wedge$-module (use Algebra, Lemma 00HC). Since

$$(\bigoplus\nolimits_{\alpha\in A} R)^\wedge = (\bigoplus\nolimits_{\alpha\in A} R^\wedge)^\wedge$$

we may replace R by $R^\wedge$ and assume that R is complete with respect to I (see Algebra, Lemma 031C). In this case Lemma 05BC tells us the map $(\bigoplus_{\alpha\in A} R)^\wedge \to \prod_{\alpha\in A} R$ is universally injective. Thus, by Algebra, Lemma 058P it suffices to show that $\prod_{\alpha\in A} R$ is flat. By Algebra, Proposition 05CZ (and Algebra, Lemma 05CY) we see that $\prod_{\alpha\in A} R$ is flat. □